

Course Code: : AI254TA

Date :

Semester : V

Time :

Max Marks : 50

Duration : 90 mins

Machine Learning Operations (MLOps)

CIE I – Scheme and Solution

Note: Answer all the Questions

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	PART – B																					
1	a Compare DevOps and MLOps. Highlight key similarities and differences in their objectives and workflows. Key Similarities (2 Marks) - Both emphasize automation, continuous integration (CI), and continuous deployment (CD). - Both encourage collaboration between development and operations teams. - Aim to improve efficiency, reliability, and scalability in production systems. Differences : Any two (2 Marks) <table><tr><th>Aspect</th><th>DevOps</th><th>MLOps</th></tr><tr><td>Focus</td><td>Application code and services</td><td>Data, models, and ML pipelines</td></tr><tr><td>Components Managed</td><td>Code repositories, build pipelines</td><td>Data pipelines, model training, versioning</td></tr><tr><td>Testing</td><td>Functional and performance testing</td><td>Model validation, bias testing, drift detection</td></tr><tr><td>Deployment</td><td>Static applications</td><td>Continuously learning models (dynamic behavior)</td></tr><tr><td>Monitoring</td><td>System uptime and logs</td><td>Model accuracy, drift, and performance metrics</td></tr></table>	Aspect	DevOps	MLOps	Focus	Application code and services	Data, models, and ML pipelines	Components Managed	Code repositories, build pipelines	Data pipelines, model training, versioning	Testing	Functional and performance testing	Model validation, bias testing, drift detection	Deployment	Static applications	Continuously learning models (dynamic behavior)	Monitoring	System uptime and logs	Model accuracy, drift, and performance metrics	4	2	1
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	b Analyze the technical and organizational challenges in implementing MLOps in large-scale enterprise environments and explain how they affect model scalability, collaboration, and governance. Answer : 📌 Technical Challenges (2 Marks) - Infrastructure Complexity: Diverse tools and environments across teams lead to inconsistent deployments and scalability issues. - Model and Data Versioning: Lack of proper tracking affects reproducibility and model governance. - Monitoring Difficulties: Limited automation in drift detection reduces performance reliability of models in production. 📌 Organizational Challenges (2 Marks) - Siloed Teams: Data scientists, DevOps, and business teams working independently hinder effective collaboration. - Weak Governance Practices: Absence of standardized workflows and documentation affects compliance and accountability. 📌 Impact (2 Mark)	6	4	3																		



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		These challenges cause delayed model deployment, duplication of effort, and lack of auditability, ultimately affecting model scalability, cross-team collaboration, and enterprise-level governance.			
	a	With a neat diagram, elaborate the process of ML Life Cycle Answer <pre> graph LR A[Problem Definition] --> B[Data Collection] B --> C[Data Cleaning & Preprocessing] C --> D[Exploratory Data Analysis (EDA)] D --> E[Feature Engineering & Selection] E --> F[Model Selection] F --> G[Model Training] G --> H[Model Evaluation and Tuning] H --> I[Model Deployment] I --> J[Model Monitoring & Maintenance] </pre> # elaborate the points in detail	6	2	1
2	b	A health-tech company is developing a machine learning system to predict patient readmission risk for hospitals. The MLOps team includes data scientists, data engineers, DevOps specialists, and a Subject Matter Expert (SME) — a senior clinician with deep medical domain knowledge. List down the role of SME in this scenario Answer: - The SME defines clear business goals such as reducing 30-day readmission rates by 15%. - They translate clinical knowledge into model features, advising which patient attributes (e.g., medication adherence, prior diagnoses, age, discharge notes) are clinically significant. - When the model performs well technically but suggests unrealistic predictions (e.g., low risk for high-severity cases), the SME identifies the mismatch between medical logic and data patterns. - The SME collaborates with data scientists to refine feature selection and evaluation metrics, ensuring the model aligns with medical standards and ethical requirements. - Post-deployment, the SME reviews model outcomes with clinicians and ensures that the feedback loop is integrated into the MLOps monitoring and retraining pipeline.	4	2	1
3	a	A national bank has deployed a Credit Risk Prediction ML model that assesses whether a customer is likely to default on a loan. Initially, the model performs well with an accuracy of 92%. However, after six months, the following issues arise: - The model begins misclassifying new types of customers due to changes in economic conditions. - Incorrect predictions start leading to financial losses and biased lending decisions.	10	4	5



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		- The operations team finds it difficult to trace which model version is running in production. - Data quality issues (missing or inconsistent customer data) cause the model's reliability to drop. Analyse the various risks associated with deploying machine learning models in a banking environment, and explain how these risks can be identified, assessed, and mitigated to ensure trust, compliance, and consistent performance." Depict the risk matrix to support your answer. Ans : (i) Risk Identification and Assessment(2 marks) (ii) Continuous Monitoring and Alerts (2 Marks) (iii) Automated Retraining and Model Versioning (1 Marks) (iv) Governance, Compliance, and Security (2 Marks) (v) Risk Matrix (2 Marks)			
4	a	Elaborate on how iteration and feedback contribute to the improvement and refinement of models in the Machine Learning life cycle. Definition/Concept and Role in ML life cycle [3 Marks] Iteration and feedback are key processes that ensure continuous improvement in the Machine Learning life cycle. - Iteration involves repeatedly training, testing, and refining the model to enhance its performance. Each cycle helps in improving data quality, tuning hyperparameters, and selecting better algorithms. - Feedback provides insights from model evaluation or user interactions to identify errors, data drift, or bias. This information guides further refinement. Example[1 Marks] In a spam detection system, incorrect classifications are reviewed and used as feedback for retraining the model, making it more accurate in subsequent iterations. Iteration and feedback help create reliable, adaptable, and high-performing ML models.	4	2	1
	b	Green Energy Insights, a renewable energy analytics firm,leverages AI models to forecast energy demand, optimize power generation, and detect equipment faults. With diverse data sources from IoT sensors and smart grids, ensuring data accuracy, transparency, and regulatory compliance became difficult. Justify how implementing a unified data governance framework would be a novel solution in the given scenario. Definition of Data Governance (1 mark)	6	4	3



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		A data governance framework is a structured approach that defines policies, roles, standards, and processes to ensure data is accurate, consistent, secure, and compliant throughout its lifecycle. Core Components (3 marks) A unified data governance framework includes: • Data Quality Management: Standardizes and validates sensor data for accuracy. • Data Lineage and Traceability: Tracks data flow from source to model for transparency. • Access Control and Security: Restricts unauthorized access and ensures data privacy. • Regulatory Compliance: Ensures adherence to energy sector regulations and audit requirements. • Metadata Management: Maintains context and ownership of every dataset. Application to Scenario (2 marks) By implementing a unified data governance framework, GreenEnergy Insights can: • Integrate heterogeneous IoT and smart grid data into a single trusted source. • Ensure data consistency across forecasting and fault detection models. • Provide audit trails for regulatory compliance and accountability. • Build transparency and trust in AI-driven energy decisions.			
5		A healthcare analytics company is building a machine learning model to predict whether a patient is at risk of developing diabetes within the next year. The dataset includes patient information such as: • Age, Gender, BMI, Blood Pressure, Glucose Level, Insulin Level, Physical Activity Score, Smoking Habits, Family History, and Diet Type. • The target variable is Diabetes Risk (Yes/No). (i) During the initial data analysis, you notice that several features have missing values, and Glucose Level has extreme outliers. How would you handle these issues during the data exploration phase, and why are these steps critical for model reliability? (ii) What new features could you create to improve model performance, and how would you justify their inclusion? (iii) After creating several new features, your dataset now has over 30 predictors. Describe the process you would follow to perform feature selection and decide which variables to keep in the final model. What methods would you use and why?	10	3, 4	2



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	(iv) Analyse how these steps influence the model's accuracy, interpretability, and generalization performance in this diabetes prediction scenario. Data Exploration -Define and explain its role with example [3 marks] Data exploration involves analyzing datasets to understand their structure, patterns, relationships, and anomalies before model training. Role: • Identifies missing values, outliers, and inconsistencies. • Helps visualize feature distributions and correlations using plots like histograms or heatmaps. • Guides the choice of algorithms and preprocessing steps. Feature Engineering - Define and describe its purpose with example [3 marks] Feature engineering is the process of creating new input features or transforming existing ones to make data more suitable for learning algorithms. Role: • Enhances the model's ability to detect patterns. • Involves scaling, encoding categorical data, handling missing values, and generating new variables. • May include transformations like normalization, log-scaling, or polynomial features. Feature Selection - Define, explain techniques, and significance [2 marks] Feature selection is the process of choosing the most relevant and significant features that contribute to model accuracy while removing redundant or noisy ones. Techniques: • Filter methods: Correlation, Chi-square test. • Wrapper methods: Recursive Feature Elimination (RFE). • Embedded methods: Lasso regression, Decision tree importance. Significance: Reduces model complexity, training time, and overfitting while improving interpretability. Impact on Model Performance - Summarize how these steps improve model efficiency and accuracy [2 marks] data exploration, feature engineering, and feature selection ensure that the input data is clean, meaningful, and optimized, resulting in: • Higher model accuracy • Better generalization to unseen data			
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		• Reduced computation cost and overfitting			
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